

Semester Project Proposal

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Instructor:	Dr. Demetris Trihinas

1. Basic Information

This project has been assigned by Dr. Demetris Trihinas as a semester project for COMP-548DL “Big Data Management and Processing”, encompassing the practical application of big data processing solution of the coursework.

The proposed title of this project is ‘**Stock Market Analysis Using Big Data Storage and Processing System**’.

This write-up aims to furnish a high-level introduction to the identified problem and proposed solution.

2. Background & Motivation

Based on my innate interest in investments and analysis of equities and similar financial instruments, I desire to apply the knowledge, tools, and techniques attained through this course in the same field.

Stock markets generate large volumes of time-series data at a continuous pace, but most free analysis workflows rely on tools like pandas or spreadsheets, which operate in-memory and do not scale as data grows. This becomes a problem when analysts want to fetch historical information for forecasting and face the challenge of storing, querying, or processing large amounts of data efficiently.

Through this project, I will strive to address this problem by designing a scalable Big Data pipeline that persistently stores stock price data and processes it using distributed tools instead of local memory.

3. Project Objectives (Hypothesis)

This project aims to understand how stock prices behave over time by analyzing both historical and intraday data through a scalable Big Data pipeline.

From the data, I want to learn:

- How price movements and volatility vary across different time windows.
- Whether aggregated indicators (e.g., moving averages, daily summaries) expose trends not visible in raw data.
- How real-time ingestion enhances analysis compared to offline historical computation.

4. Data Sources

The main dataset for this project will be collected from Alpha Vantage API, which provides historical and intraday stock price time-series data in JSON format. Data will be ingested through Python scripts. Alternatively, two other sources Twelve Data API (provides a similar JSON time-series structure) and Yahoo Finance unofficial API will be availed if required.

The project will be addressing mainly two dimensions of big data dimensions namely volume and velocity.

5. Big Data Dimensions

This project reflects a Big Data problem because the dataset increases continuously over time (**Volume**) and stock prices arrive every minute (**Velocity**). These dimensions make traditional in-memory analysis impractical and require scalable storage and distributed processing to extract actionable insights.

6. Solution Overview

This project will address the problem by designing and implementing a scalable Big Data architecture that ingests, stores, and processes stock market time-series data using concepts covered in our course.

1. Data from Alpha Vantage / Twelve Data API / Yahoo Finance will be collected through automated API calls, converted from JSON into a structured time-series schema, and stored persistently in a distributed NoSQL database (MongoDB Atlas).
2. To process the data efficiently at scale, analytical workloads such as moving averages, volatility calculation, aggregations, and intraday summaries will be executed using Apache Spark batch jobs, following the Big Data programming models introduced in the course such as MapReduce-style transformations.

7. Tools/libraries

The following tools and libraries are currently envisaged (however, this list is not exhaustive and may be updated over the project lifecycle)

1. Python for data ingestion
2. Alpha Vantage API / Twelve Data API / Yahoo Finance as data providers
3. MongoDB Atlas for Data Storage
4. Spark for big date processing
5. Jupyter Notebook for visualization of results
6. Github as repository of codes